

Course Objectives

The students will be exposed to the fundamentals of machine learning: classification, regression, supervised and unsupervised learning. They will learn which algorithms to use in which context, including model validation and evaluation. They will be exposed to a thorough survey of the fundamental security applications that machine learning provides as well the current security limitations of machine learning.

Course Outcomes

COs	Description
CO1	Learn and understand what machine learning and artificial intelligence is
CO2	Gain proficiency in scikit-learn, using supervised and unsupervised learning
CO3	Learn the fundamentals of regression and classification
CO4	Make use of classification and anomaly detection systems in security – fraud and spam detection
CO5	Learn to threat model for machine learning, understanding adversarial attacks

CO-PO Mapping

PO/PS O CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	3	2	-	-	-	3
CO2	1	2	3	-	-	-	3
CO3	3	3	2	-	-	-	3
CO4	2	3	3	-	-	-	3
CO5	2	2	3	-	-	-	3
CAM	2	3	3	-	-	-	3

Syllabus

Python, Jupyter Notebooks, Pandas, Numpy, Matplotlib, Seaborn, Scikit-Learn. Mathematics review: derivatives, gradients, sums, products. Supervised learning: Linear regression, Decision Trees, Support Vector Machines, K-nearest neighbors, random forests, adaboost, gradient boosting,. Unsupervised learning: k-means, clustering, dbscan, , PCA, Bias-variance tradeoff. Learning and validation curves. Cross validation, shuffle split, k-fold, Random seeds. Baseline and benchmarking models. Gradient descent, regularization, feature scaling, one hot encoding, label encoding. Train-test-split. Metrics: accuracy, f1-score, precision, recall, confusion matrices. Gini impurity, information gain ratio,. Hyper-parameter tuning with gridsearch and random search Natural language processing, ngrams, bag of words, vectorizers. Pipelines in scikit-learn to avoid overfitting. Data wrangling with feature preprocessing and EDA. Machine learning for security - anomaly detection, fraud detection, malware detection, spam detection, phishing detection,. Security of machine learning: adversarial attacks on machine learning. Data poisoning, model stealing, evasion attacks at inference time

Textbooks / References:

1. Tom M Mitchell, Machine Learning, McGraw Hill, 1997
2. Jake Vanderplas, Python Data Science Handbook, O'Reilly Media, 2016
3. Clarence Chio, David Freeman, Machine Learning and Security, O'Reilly Media, 2018

Evaluation plan

Internal	Quizzes 30%	30%	100%
External	MCQ's +Descriptive – 50 Marks	70%	
	Final Exam Lab MCQ section – 20 Marks		

AMRITA VISHWA VIDYAPEETHAM Amrita Online - Course Plan					
Week	Topic	Learning Objectives	eLearning Content	Video Lectures	Assessments
1	Introduction to AI	Python Jupyter Notebooks to create and share your documentation Pandas to perform transformations and calculations of datasets Numpy to perform numerical computing Matplotlib and Seaborn for Data Visualization Scikit-Learn for machine learning LAB: Practice with the Tool	8 videos 1 quiz	Welcome to the course Learning from patterns What is AI How do computers learn? Scikit-Learn and Jupyter Numpy Matplotlib Pandas	Quiz 1
2	Math Review	Derivatives Sums	2 videos 1 quiz + 1 lab quiz	Math review Linear intuitions	Quiz 2 + lab quiz 1

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Amrita Online - Course Plan

Week	Topic	Learning Objectives	eLearning Content	Video Lectures	Assessments
3	Gradient descent	Linear Regression Intuitions Bias Variance Tradeoff (overfitting vs underfitting) LAB: Linear Regression in 2D Gradient Descent Regularization Feature Scaling LAB: Gradient Descent and Regularization in 2D LAB : Predicting Housing Prices	7 videos 1 quiz + 2 lab quiz	Gradient descent part 1 Gradient descent part 2 Gradient descent part 3 Gradient descent part 4 Regularization Basics of a Dataset Feature scaling	Quiz 3 + lab quiz 2 and 3
4	Evaluation and train	Model Evaluation and Validation I Training and Test Sets Baseline Models Random Seeds LAB : Predicting Housing Prices Extended Introduction to	3 videos 1 quiz + 1 lab quiz	Evaluation: Test Sets Train sets in scikit-Learn Baseline Models	Quiz 4 + lab quiz 4
5	Model Evaluation and Validation I	Classification Metrics Accuracy Precision Recall F1-score Confusion Matrices Classification Reports LAB: Detecting breast cancer	5 videos 1 quiz + 1 lab quiz	Classification Metrics Accuracy and confusion matrices Precision, recall and F-1 score Choosing a metric	Quiz 5 + lab quiz 5
6	Classification with Decision Trees	Decision Trees Gini Impurity Coefficient Information Gain Ratio Regularization parameters LAB: Classifying Spam with Decision Trees	2 videos 1 quiz + 1 lab quiz	Decision trees Splitting mechanisms	Quiz 6 + lab quiz 6

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Amrita Online - Course Plan

Week	Topic	Learning Objectives	eLearning Content	Video Lectures	Assessments
7	Model Evaluation and Validation II	Hyperparameter Tuning Gridsearch Randomized Search K-fold, ShuffleSpit, Leave-One-Out, Stratification Learning and Validation Curves LAB: Validation and Evaluation of your Spam model.	4 videos 1 quiz + 1 lab quiz	Validation schemes Searching for Hyperparameters Learning curves Validation curves	Quiz 7 + lab quiz 7
8	Ensemble Methods	Bagging with Random Forests Boosting with Adaboost LAB: Classifying handwritten digits with Ensembles	3 videos 1 quiz + 1 lab quiz	Bagging Random forests Boosting	Quiz 8 + lab quiz 8
9	Dealing with Data	Feature extraction from different data types Tabular Images Time Series Natural Language Processing Computer vision Handling missing data Data pipelines to avoid overfitting Lab: Wrangle a dataset from UCI Lab: NLP pipelines with the 20Newsgroup Dataset	5 videos 1 quiz + 2 lab quiz	Intro to dealing with data Tabular data Vision and sound Natural language processing Pipelines	Quiz 9 + lab quiz 9 and 10
10	Unsupervised Learning	Clustering K-means DBscan LAB: Clustering on 2D blobs	5 videos 1 quiz + 2 lab quiz	Unsupervised learning - What is it for? Introduction to clustering K-means clustering Db-scan Clustering II	Quiz 10 + lab quiz 11 and 12

AMRITA VISHWA VIDYAPEETHAM Amrita Online - Course Plan					
Week	Topic	Learning Objectives	eLearning Content	Video Lectures	Assessments
11	Unsupervised Learning	Dimensionality Reduction The curse of dimensionality PCA LAB: Clustering and PCA on Retail Customer Data	2 videos 1 quiz + 1 lab quiz	The curse of Dimensionality PCA	Quiz 11 + lab quiz 13
12	Machine Learning for Security	Anomaly detection Malware detection Network intrusion detection Phishing detection	4 videos 1 quiz	ML for Cybersecurity Can ML help? Phishing detection Finding Datasets	Quiz 12
13	Security of Machine Learning	Adversarial attacks Vision Phishing	5 videos 1 quiz	Adversarial Evasion attacks Intuition of evasion attacks Threat modelling evasion attacks Black box attacks Heterogeneous vs homogeneous datasets	Quiz 13
14	Security of Machine Learning	Threat Models for machine learning Methods in which machine learning systems fail (what are the security risks)	6 videos 1 quiz	Points of failure Defense against attack Industry readiness Governance, risk and compliance for ML Penetration testing ML Thank you and good bye!	Quiz 14

Faculty Information

1. Mr. Cherukupalli Veda Vyasa Aditya
2. Ms. Haritha